

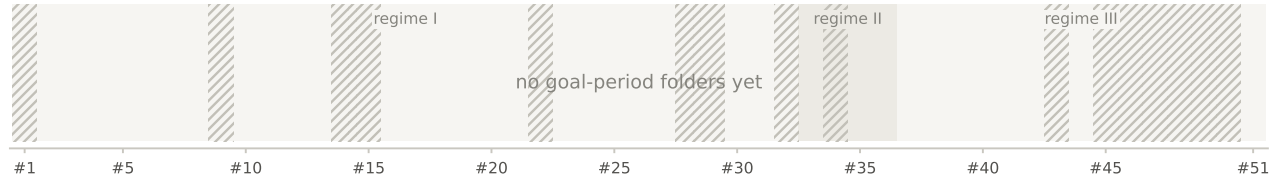
H71: Memory is a first-order homeostat

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

not yet rated

Bottom line. *Pending.*

level: hypothesis · pipeline checks 14% · updated —



Goal periods tested (none yet). ✔ supported ✖ failed ~ mixed d descriptive ... pending. Heavy outline: confirmatory test on the locked holdout. Hatched columns: held-out goal periods. Brackets: tests spanning a natural experiment.

WHAT HAPPENED

Not yet summarized. The question: Do deviations of memory size from its set point relax at a fixed per-agent rate, with overshoot after consolidations and erasures and a gain change at NE41, so a first-order control model beats a random walk?

PHYSICAL MODEL

From: physics-models/*i*NN-slugs/ *i*Which variant of that model this hypothesis uses, its parameters, and what it predicts here. E.g. "SIS with spontaneous adoption field ϵ ; λ , γ , ϵ fit per regime." *i*

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Pending.

Observables. *Pending.*

RESULTS VS PREDICTIONS

Pending.

GOAL PERIODS IN DETAIL

No goal-period folders yet.

FAITHFULNESS SCORECARD

A mapping	–	F identifiability	–
B assumptions	–	G ground truth	–
C adequacy	–	H comparative	–
D unfitted predictions	–	I transfer	–
E interventional	–		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Pending.

ROUND 2

No round-2 redirects yet.

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fra/summaries/build-summaries.py · card hypotheses/H71-memory-
homeostat/README.md · ratings: not rated · DRAFT: auto stub, not
yet written (page 2)